Should a Canadian Province Specialize More?

Scale, uncertainty, diversification and worker protection

Prepared for the self-directed week of November 9, 2026. Work in pairs.

Deadline: the instructor will confirm the submission date on Canvas.

The decision you must answer

Choose one Canadian province and one clearly defined trade-exposed industry or cluster. Advise its economic-development agency: should it deepen an existing specialization, develop a feasible adjacent activity, or preserve specialization while improving workers' protection against shocks? Compare the province with two other provinces and Canada. Make one conditional recommendation backed by reproducible evidence, and state what would change your judgment.

Why this question follows our course

Comparative advantage explains reasons to specialize. Krugman explains gains from scale and variety. H–O and Stolper–Samuelson teach us to ask who receives the gains. Ma and Ramondo add a question: what if workers choose an industry and location before knowing which shocks will arrive, and cannot adjust instantly? Openness can support specialization and also provide alternative customers and suppliers. The task is to preserve useful gains while understanding exposure.

This is a Canadian evidence exercise. You are not expected to estimate the paper's structural model, solve a national optimum, or reproduce its U.S. welfare numbers for Canada. A well-supported finding can be conditional or ambiguous. Neither maximum specialization nor maximum diversification is a prescribed answer.

Your output: a professional decision document

1. **One memo PDF: exactly four narrative pages**, with two principal exhibits inside those pages. Use 11-point text, ordinary margins and no cover page. Add at most one references/data page and one reproducibility/AI-use page.
2. **One research ZIP:** README and run command, unchanged starter/source files, processed data, code, figures, source audit, verification report, AI log and both partners' contributions. One partner submits both files for the pair; put both full Canvas names on the memo and in the README.

Workflow. Professional messaging, shared files and Zoom are suitable; physical meetings are not required. Both partners must understand the analysis.

Research-assistant standard

I am treating you as research assistants. AI can help obtain data, write code and challenge a draft. You supply the judgment: does the evidence answer the question, are the numbers right, and could the memo responsibly be circulated to a policy reader? Compare your reasoning and presentation with the anonymous authentic student example.

Understand the Paper's Mechanism

Read the current version, not a mixture of drafts

Xiao Ma and Natalia Ramondo, *Comparative Advantage and Economies of Scale under Uncertainty*, NBER Working Paper 35942, October 2026. Start with the introduction and conclusion; inspect Table 6 (printed p. 45, PDF p. 47), Table 7 (printed p. 46), and the insurance comparison in Table 10 (printed p. 51). The course podcast supplies an undergraduate explanation.

The supplied 2025 PHBS draft predates this revision. The latest Table 6 reports U.S. gains of 1.87% with uncertainty versus 2.27% without it: 0.40 percentage points, or about 18% less relative to 2.27%. These are calibrated consumption-equivalent welfare comparisons with autarky, not Canadian estimates, GDP growth rates or causal treatment effects. The revised baseline has positive gains for every U.S. commuting zone; some no-scale variants have regional losses.

Five links to explain in plain English

1. **Commitment before the shock:** people select a location and industry before uncertainty is resolved. Skills, housing and networks make instant switching an unrealistic benchmark. The paper fixes the choice after the shock.
2. **Risk matters for well-being:** a high average payoff can be less attractive if bad outcomes are severe and workers are poorly insured.
3. **Local external scale:** a larger local industry can support a more productive cluster. This is different from one Krugman firm's fixed cost being spread over more units. A big industry alone is not evidence of increasing returns.
4. **Trade works in both directions:** specialization and foreign shocks can increase exposure; alternative sources of supply and demand can diversify it. The net effect depends on what shocks move together and on prices and wages.
5. **Institutions change the trade-off:** insurance and adjustment possibilities can protect workers without discarding all specialization gains. Idealized insurance in a model does not establish a real program's effectiveness.

A small numerical illustration, not a Canadian estimate

Suppose two equally likely states give real consumption of 120 or 60 under a specialized plan, versus 84 in either state under a stable plan. Their averages are 90 and 84. For the illustrative utility $u(c) = -1/c$, the certain amount with the same expected utility is

$$CE = \frac{1}{0.5/c_{\text{good}} + 0.5/c_{\text{bad}}}.$$

The specialized plan has $CE = 80$, below the stable plan's 84. An *assumed* 10% consumption improvement in both specialized states gives 132 and 66, and $CE = 88$, reversing the comparison. Scale benefits and risk both matter. These payoffs and the 10% improvement are invented, not estimates from the paper.

A Feasible Canadian Analysis

Starter: `canada_specialization_starter.zip`, supplied with this brief. It contains a small course extract from Statistics Canada, not completed answers. Preserve the files unchanged and read the data dictionary first.

1. Fix your scope and establish actual trade exposure

Choose a province, two comparators, and a cluster: for example, forest products in BC, energy in Alberta, autos in Ontario, or another justified case. Use **2010–2024** as the common window. Supply at least one independent official fact establishing the cluster's trade exposure and one source about skills, suppliers, infrastructure or production capabilities. Do not assume employment in broad manufacturing measures auto employment, or that the combined resource group measures forestry alone. Public services can affect employment stability without being export industries.

2. Describe the industrial mix

The starter has annual employment in *thousands of persons* for 16 non-overlapping industry groups, Canada and ten provinces. With E_{pjt} denoting employment in province p, industry j, year t, compute

$$s_{pjt} = \frac{E_{pjt}}{\sum_{k=1}^{16} E_{pkt}}, \qquad HHI_{pt} = \sum_{j=1}^{16} s_{pjt}^2, \qquad LQ_{pjt} = \frac{s_{pjt}}{s_{\text{Canada},jt}}.$$

Report 2010 and 2024 shares and HHI. Explain $LQ > 1$ as an above-national *employment share*; it is not export RCA or a measured opportunity cost. Reconcile the included sum with the separate all-industry total; allow documented rounding differences. Do not combine total/parent sectors with their components.

3. Measure historical variation, not the variance of a trend

For positive employment levels, compute annual percentage growth for 2011–2024:

$$g_{pjt} = 100 \left(\frac{E_{pjt}}{E_{pj,t-1}} - 1 \right).$$

Report each industry's mean and *sample* standard deviation ($n - 1$ denominator, percentage points). There are only 14 annual growth observations. These mix demand, supply, productivity, policy and survey noise: they do not identify the paper's productivity shocks or future uncertainty. Mean employment growth is not a productivity or welfare payoff.

4. Account for co-movement with one simple weighted series

Hold the 2010 employment shares fixed and calculate, for every year,

$$G_{pt} = \sum_{j=1}^{16} s_{pj,2010}\, g_{pjt}.$$

Then take the mean and sample SD of the resulting annual series. This captures historical co-movement; averaging industry SDs does not. G is a fixed-mix accounting screen, not exactly observed aggregate growth or a job forecast.

Build a Decision from the Evidence

5. Test one explicitly hypothetical change in the mix

Select an industry A from which to reduce weight and a plausible adjacent industry B to receive it. Use a five-percentage-point change unless this would remove more than half of A's initial share:

$$\delta = \min(0.05, s_{pA,2010}/2), \qquad s'_A = s_A - \delta, \quad s'_B = s_B + \delta.$$

Keep all other shares unchanged. Check nonnegativity and a sum of one. Reuse the same historical industry growth observations, recompute G'_t and its mean/SD, and compare them with the original fixed mix. Test a half-sized change $\delta/2$ as well. Explain why B is feasible using capabilities evidence, rather than simply selecting the least volatile industry after seeing the data.

This is not an optimal allocation. It holds industry growth rates fixed; moving real workers could change productivity, wages, prices and local scale. A lower SD need not improve welfare, especially if productive scale is lost.

Two principal exhibits, each carrying a claim

1. **Where the province is concentrated.** A clear share or LQ comparison with the two provinces and Canada, accompanied by 2010/2024 concentration values. Use a focused heatmap or bar chart; do not overwhelm the reader with all cells.
2. **What diversification would and would not insure.** A plot or compact panel showing industry mean employment growth versus historical SD, plus the baseline and reweighted fixed-mix results. Include the smaller-shift sensitivity and a visible check on whether exceptional years drive the result.

Repeat the variability calculation excluding the *2020 and 2021 growth observations*. Do not create a new 2019–2022 growth rate. Report the remaining 12 observations and any change in the ranking. This is sensitivity, not a claim that removing those years identifies the world without a pandemic.

Give the four pages a clear job

1	Finding first: decision, province/cluster, baseline, recommendation and confidence.
2	Mechanism: specialization benefit, local scale, shock, commitment and who bears risk.
3	Two exhibits: interpret magnitudes and the strongest contrary evidence.
4	Decision: policy design, costs/beneficiaries, sensitivity and observable reversal condition.

A responsible recommendation

Choose a targeted action, such as supplier development, export-market diversification, retraining or a defined insurance/adjustment intervention. Explain its funding or implementation constraint. Employment data alone cannot rank its welfare impact; say which additional evidence is needed. Do not infer that closing trade is an effective hedge without evaluating what openness insures.

Match Your Claim to the Evidence

For the final project. This new paper and the companion causal-reasoning workshop will not be tested on Midterm 2. The slides assume no previous knowledge of causal inference and explain one possible comparison step by step.

What kind of claim are you making?

- **Description:** "Employment fell after the tariff." Audited data can establish this pattern. Other events may also explain the fall.
- **Causation:** "The tariff caused employment to fall." This asks how employment compares with what would have happened to the same group, at the same time, without the tariff. That unobserved alternative is the *counterfactual*.
- **Conditional scenario:** "Holding each industry's observed historical growth rates unchanged, this alternative mix has lower historical variability." State the assumption. Actual policy could change wages, prices, growth rates and local scale.

When your recommendation needs causal support

If your recommendation depends on X causing Y, support that link with a credible comparison or a relevant existing causal study. Explain the study's assumptions and whether its workers, sector, country and policy match your case. A before-and-after change, a regression, a prediction or an economic mechanism alone does not supply the missing comparison.

Differences-in-differences (DiD) is one possible tool: compare the change in an affected group with the change in a suitable comparison group. Its key assumption is that, without the policy, the groups would have had the same average change. Similar earlier trends help assess this assumption but cannot prove it. Other shocks, early responses, supply-chain effects and changing measurement can make the comparison misleading.

Choose a feasible and honest claim

You may need causal tools to support a causal policy claim. You are not required to run DiD or produce an original causal estimate for every project. Relevant published causal evidence is one route. Descriptive evidence and clearly labelled scenarios remain useful when you narrow the claim and state what remains unknown. This guidance adds no new deliverable or grading criterion.

Companion workshop: https://vaguiarl.github.io/econ342-codex-labs/final-project-tools/

Sources, Codex Workflow and Verification

Primary sources: clickable and copyable

- **Paper:** https://www.nber.org/papers/w35942
- **Employment table:** https://doi.org/10.25318/1410002301-eng
- **Full employment download:** https://www150.statcan.gc.ca/n1/tbl/csv/14100023-eng.zip
- **Provincial merchandise trade:** https://doi.org/10.25318/1210017501-eng
- **Optional real GDP by industry:** https://doi.org/10.25318/3610071101-eng

The employment starter covers 2000–2025 for optional checks; the common analysis ends in 2024. It has a national series plus ten provinces, not territories. Trade data are goods only: province of origin, domestic versus total exports, partner overlap and nominal prices require care. Chained real GDP is useful for growth, but its components are not additive and must not supply nominal shares.

Use separate AI roles or successive critical passes

1. **Planner:** define the decision, groups, shock and competing mechanisms.
2. **Data analyst:** inspect actual columns and flags; write reproducible transformations; create the two exhibits from unchanged source files.
3. **Skeptical reviewer:** check equations, units and claims; try to overturn the recommendation with the sensitivity and contrary evidence.
4. **Writer/editor:** produce a four-page decision narrative after the analysis. Both students personally verify the claims and rerun the code.

Copyable prompts

Start: "Read the brief and data dictionary. Do not write the memo yet. List the decision, theory, baseline, measures and possible refutations. Inspect the files before suggesting columns or joins. Do not invent data."
Compute: "Use the 16 non-overlapping groups. Keep units and flags. Calculate shares, HHI, LQ, annual growth and the fixed-weight growth series. Take SD after weighting annual growth. Validate every identity and report sample size."
Challenge: "Which claims confuse employment with productivity, welfare, export RCA or increasing returns? What co-movement, scale loss or unmeasured cost could reverse the recommendation? Separate data, scenario and causal claim."

Verification report: unique province–industry–year keys; missingness and source flags; three raw cells checked manually; one hand-calculated share, growth rate and SD; totals/rounding reconciliation; weights nonnegative and summing to one; no parent/child double counting; exact sample/window; unchanged sources and run command; every figure regenerated. Preserve both AI errors and your corrections in the AI log. Private credentials never belong in the ZIP.

Rubric and Submission Check

Diagnostic quality: four criteria, each 0–5

The professional bar is the anonymous student benchmark, adapted to this question. A complete submission can still have weak analysis. The real criterion scores and concrete feedback remain visible even when completion credit is full.

Criterion	5: excellent	Around 3: developing
Decision and thesis	Directly answers the defined Canadian decision; defends a conditional recommendation against a serious alternative.	Relevant topic and plausible conclusion, but baseline, decision or reversal rule is weak.
Economic mechanism	Correctly links specialization, local external scale, uncertainty, commitment and heterogeneous workers; explains insurance and trade's two-sided role.	Names the forces but does not fully trace their interaction or confuses local external scale with firm fixed costs.
Evidence and verification	Audited real data, valid denominators, covariance-aware comparison, informative exhibits and sensitivity; claims match what the design identifies.	Mostly sound calculations but a fragile comparison, incomplete check, weak trade link or overstated inference.
Professional readiness	A skeptical policy reader can understand, reproduce and responsibly circulate the memo; contrary evidence, implementation and uncertainty are treated seriously.	Readable and partly reproducible, but needs substantive checking or rewriting before circulation.

Common scale: 5 = excellent; 4 = strong with bounded repairs; 3 = developing with material gaps; 2 = weak; 1 = insufficient; 0 = criterion absent or not demonstrated. The diagnostic total is out of 20. A policy-ready score is earned by judgment and evidence, not by polished prose alone.

Canvas course credit is separate

- **100:** fully satisfies the assigned task and required deliverables.
- **50:** medium/partial compliance, such as an incomplete empirical comparison, a materially incorrect central calculation, or a missing required package.
- **0:** no delivery. An inaccessible file or unresolved partner match is held for review rather than assumed to be no delivery.

Diagnostic points are never converted mechanically into 0/50/100. Full completion credit does not mean the memo is excellent or ready for Ottawa.

Before you submit

Does the first paragraph answer the question? Do the figures support that answer? Have you shown a meaningful counterargument? Can your partner reproduce every number? Are an observed fact, a historical screen, a scenario and a recommendation clearly distinguished? Upload the memo PDF and research ZIP, then reopen both.